Model for "N198R"

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General Condition

- Basis type: lg
- SAMB selection:
  - Type: [Q, G]
  - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
  - Irrep.: [A]
  - Spin (s): [0, 1]
- Atomic selection:
  - Type: [Q, G, M, T]
  - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
  - Irrep.: [A, E, T]
  - Spin (s): [0, 1]
- Site-cluster selection:
  - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
  - Irrep.: [A, E, T]
- Bond-cluster selection:
  - Type: [Q, G, M, T]
  - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
  - Irrep.: [A, E, T]
- Max. neighbor: 10
- Search cell range: (-2, 3), (-2, 3), (-2, 3)
- Toroidal priority: false

Group and Unit Cell

- Group: SG No. 198 T^4 $P2_13$ [cubic]
- Associated point group: PG No. 198 T 23 [cubic]
- Unit cell:
  $a =$ 1.00000, $b =$ 1.00000, $c =$ 1.00000, $\alpha =$ 90.0, $\beta =$ 90.0, $\gamma =$ 90.0
- Lattice vectors (conventional cell):
  $\boldsymbol{a}_1 =$ [1.00000, 0.00000, 0.00000]
  $\boldsymbol{a}_2 =$ [0.00000, 1.00000, 0.00000]
  $\boldsymbol{a}_3 =$ [0.00000, 0.00000, 1.00000]

Symmetry Operation

Table 1: Symmetry operation

	#	SO	#	SO	#	SO	#	SO	#	SO
	1	$\{1\|0\}$	2	$\{2_{001}\|\frac{1}{2}0\frac{1}{2}\}$	3	$\{2_{010}\|0\frac{1}{2}\frac{1}{2}\}$	4	$\{2_{100}\|\frac{1}{2}\frac{1}{2}0\}$	5	$\{3^{+}_{111}\|0\}$
	6	$\{3^{+}_{-11-1}\|\frac{1}{2}\frac{1}{2}0\}$	7	$\{3^{+}_{1-1-1}\|\frac{1}{2}0\frac{1}{2}\}$	8	$\{3^{+}_{-1-11}\|0\frac{1}{2}\frac{1}{2}\}$	9	$\{3^{-}_{111}\|0\}$	10	$\{3^{-}_{1-1-1}\|0\frac{1}{2}\frac{1}{2}\}$
	11	$\{3^{-}_{-1-11}\|\frac{1}{2}\frac{1}{2}0\}$	12	$\{3^{-}_{-11-1}\|\frac{1}{2}0\frac{1}{2}\}$						

Harmonics

Table 2: Harmonics

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
1	$\mathbb{G}_0(A)$	A	0	G, M	-	-	1
2	$\mathbb{Q}_0(A)$	A	0	Q, T	-	-	1
3	$\mathbb{G}_3(A)$	A	3	G, M	-	-	$\sqrt{15}xyz$
4	$\mathbb{Q}_3(A)$	A	3	Q, T	-	-	$\sqrt{15}xyz$
5	$\mathbb{Q}_4(A)$	A	4	Q, T	-	-	$\frac{\sqrt{21}\left(x^4-3x^2y^2-3x^2z^2+y^4-3y^2z^2+z^4\right)}{6}$
6	$\mathbb{Q}_{2,1}(E)$	E	2	Q, T	-	1	$\frac{-x^2-y^2+2z^2}{2}$
7	$\mathbb{Q}_{2,2}(E)$					2	$\frac{\sqrt{3}(x^2-y^2)}{2}$
8	$\mathbb{G}_{1,1}(T)$	T	1	G, M	-	1	x

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
9	$\mathbb{G}_{1,2}(T)$					2	y
10	$\mathbb{G}_{1,3}(T)$					3	z
11	$\mathbb{Q}_{1,1}(T)$	T	1	Q, T	-	1	x
12	$\mathbb{Q}_{1,2}(T)$					2	y
13	$\mathbb{Q}_{1,3}(T)$					3	z
14	$\mathbb{Q}_{2,1}(T)$	T	2	Q, T	-	1	$\sqrt{3}yz$
15	$\mathbb{Q}_{2,2}(T)$					2	$\sqrt{3}xz$
16	$\mathbb{Q}_{2,3}(T)$					3	$\sqrt{3}xy$
17	$\mathbb{Q}_{3,1}(T,1)$	T	3	Q, T	1	1	$\frac{x\left(2x^2-3y^2-3z^2\right)}{2}$
18	$\mathbb{Q}_{3,2}(T,1)$					2	$\frac{y\left(-3x^2+2y^2-3z^2\right)}{2}$
19	$\mathbb{Q}_{3,3}(T,1)$					3	$\frac{z\left(-3x^2-3y^2+2z^2\right)}{2}$

Basis in full matrix

Table 3: dimension = 16

#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)
0	$\|s\rangle$ @Co(1)	1	$\|p_x\rangle$ @Co(1)	2	$\|p_y\rangle$ @Co(1)	3	$\|p_z\rangle$ @Co(1)	4	$\|s\rangle$ @Co(2)
5	$\|p_x\rangle$ @Co(2)	6	$\|p_y\rangle$ @Co(2)	7	$\|p_z\rangle$ @Co(2)	8	$\|s\rangle$ @Co(3)	9	$\|p_x\rangle$ @Co(3)
10	$\|p_y\rangle$ @Co(3)	11	$\|p_z\rangle$ @Co(3)	12	$\|s\rangle$ @Co(4)	13	$\|p_x\rangle$ @Co(4)	14	$\|p_y\rangle$ @Co(4)

continued ...

Table 3

	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)
	15	$\|p_z\rangle$ @Co(4)								

Table 4: Atomic basis (orbital part only)

orbital	definition
$\|s\rangle$	1
$\|p_x\rangle$	x
$\|p_y\rangle$	y
$\|p_z\rangle$	z

SAMB: 20 (all 20)

- `Co` : 'Co' site-cluster
  * bra: $\langle s|$
  * ket: $|s\rangle$
  * wyckoff: `4a`

  z1 $\quad \mathbb{Q}_0^{(c)}(A) = \mathbb{Q}_0^{(a)}(A)\mathbb{Q}_0^{(s)}(A)$

- `Co` : 'Co' site-cluster
  * bra: $\langle s|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: `4a`

z2 $\mathbb{Q}_0^{(c)}(A) = \dfrac{\sqrt{3}\left(\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(s)}(T) + \mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(s)}(T) + \mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(s)}(T)\right)}{3}$

- `Co` : 'Co' site-cluster
  * bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: `4a`

  z3 $\mathbb{Q}_0^{(c)}(A) = \mathbb{Q}_0^{(a)}(A)\mathbb{Q}_0^{(s)}(A)$

  z4 $\mathbb{Q}_3^{(c)}(A) = \dfrac{\sqrt{3}\left(\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(s)}(T) + \mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(s)}(T) + \mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(s)}(T)\right)}{3}$

- `Co;Co_001_1` : 'Co'-'Co' bond-cluster
  * bra: $\langle s|$
  * ket: $|s\rangle$
  * wyckoff: `12a@12b`

  z5 $\mathbb{Q}_0^{(c)}(A) = \mathbb{Q}_0^{(a)}(A)\mathbb{Q}_0^{(b)}(A)$

- `Co;Co_001_1` : 'Co'-'Co' bond-cluster
  * bra: $\langle s|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: `12a@12b`

  z6 $\mathbb{Q}_0^{(c)}(A,a) = \dfrac{\sqrt{3}\left(\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T) + \mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T) + \mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)\right)}{3}$

  z7 $\mathbb{Q}_0^{(c)}(A,b) = \dfrac{\sqrt{3}\left(\mathbb{T}_{1,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,a) + \mathbb{T}_{1,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,a) + \mathbb{T}_{1,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,a)\right)}{3}$

  z8 $\mathbb{Q}_0^{(c)}(A,c) = \dfrac{\sqrt{3}\left(\mathbb{T}_{1,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,b) + \mathbb{T}_{1,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,b) + \mathbb{T}_{1,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,b)\right)}{3}$

  z9 $\mathbb{Q}_3^{(c)}(A) = \dfrac{\sqrt{3}\left(\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T) + \mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T) + \mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)\right)}{3}$

  z10 $\mathbb{Q}_4^{(c)}(A) = \dfrac{\sqrt{3}\left(\mathbb{Q}_{1,1}^{(a)}(T)\mathbb{Q}_{3,1}^{(b)}(T,1) + \mathbb{Q}_{1,2}^{(a)}(T)\mathbb{Q}_{3,2}^{(b)}(T,1) + \mathbb{Q}_{1,3}^{(a)}(T)\mathbb{Q}_{3,3}^{(b)}(T,1)\right)}{3}$

z11 $\mathbb{G}_{0}^{(c)}(A) = \dfrac{\sqrt{3}\left(\mathbb{T}_{1,1}^{(a)}(T)\mathbb{M}_{1,1}^{(b)}(T) + \mathbb{T}_{1,2}^{(a)}(T)\mathbb{M}_{1,2}^{(b)}(T) + \mathbb{T}_{1,3}^{(a)}(T)\mathbb{M}_{1,3}^{(b)}(T)\right)}{3}$

- `Co;Co_001_1` : 'Co'-'Co' bond-cluster
  * bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: `12a@12b`

z12 $\mathbb{Q}_{0}^{(c)}(A,a) = \mathbb{Q}_{0}^{(a)}(A)\mathbb{Q}_{0}^{(b)}(A)$

z13 $\mathbb{Q}_{0}^{(c)}(A,b) = \dfrac{\sqrt{5}\left(\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) + \mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T) + \mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) + \mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T) + \mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)\right)}{5}$

z14 $\mathbb{Q}_{0}^{(c)}(A,c) = \dfrac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(a)}(T)\mathbb{M}_{1,1}^{(b)}(T) + \mathbb{M}_{1,2}^{(a)}(T)\mathbb{M}_{1,2}^{(b)}(T) + \mathbb{M}_{1,3}^{(a)}(T)\mathbb{M}_{1,3}^{(b)}(T)\right)}{3}$

z15 $\mathbb{Q}_{3}^{(c)}(A,a) = \dfrac{\sqrt{3}\left(\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{1,1}^{(b)}(T) + \mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{1,2}^{(b)}(T) + \mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{1,3}^{(b)}(T)\right)}{3}$

z16 $\mathbb{Q}_{3}^{(c)}(A,b) = -\dfrac{\sqrt{3}\left(\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{3,1}^{(b)}(T,1) + \mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{3,2}^{(b)}(T,1) + \mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{3,3}^{(b)}(T,1)\right)}{3}$

z17 $\mathbb{Q}_{4}^{(c)}(A) = \dfrac{\sqrt{30}\left(3\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E) - 2\mathbb{Q}_{2,1}^{(a)}(T)\mathbb{Q}_{2,1}^{(b)}(T) + 3\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) - 2\mathbb{Q}_{2,2}^{(a)}(T)\mathbb{Q}_{2,2}^{(b)}(T) - 2\mathbb{Q}_{2,3}^{(a)}(T)\mathbb{Q}_{2,3}^{(b)}(T)\right)}{30}$

z18 $\mathbb{G}_{0}^{(c)}(A,a) = \dfrac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,a) + \mathbb{M}_{1,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,a) + \mathbb{M}_{1,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,a)\right)}{3}$

z19 $\mathbb{G}_{0}^{(c)}(A,b) = \dfrac{\sqrt{3}\left(\mathbb{M}_{1,1}^{(a)}(T)\mathbb{T}_{1,1}^{(b)}(T,b) + \mathbb{M}_{1,2}^{(a)}(T)\mathbb{T}_{1,2}^{(b)}(T,b) + \mathbb{M}_{1,3}^{(a)}(T)\mathbb{T}_{1,3}^{(b)}(T,b)\right)}{3}$

z20 $\mathbb{G}_{3}^{(c)}(A) = \dfrac{\sqrt{2}\left(\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{2,2}^{(b)}(E) - \mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{2,1}^{(b)}(E)\right)}{2}$

- bra: $\langle s|$
- ket: $|s\rangle$

x1 $\quad \mathbb{Q}_0^{(a)}(A) = \begin{bmatrix} 1 \end{bmatrix}$

- bra: $\langle s|$
- ket: $|p_x\rangle\,,\ |p_y\rangle\,,\ |p_z\rangle$

x2 $\quad \mathbb{Q}_{1,1}^{(a)}(T) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & 0 \end{bmatrix}$

x3 $\quad \mathbb{Q}_{1,2}^{(a)}(T) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$

x4 $\quad \mathbb{Q}_{1,3}^{(a)}(T) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$

x5 $\quad \mathbb{T}_{1,1}^{(a)}(T) = \begin{bmatrix} \frac{\sqrt{2}i}{2} & 0 & 0 \end{bmatrix}$

x6 $\quad \mathbb{T}_{1,2}^{(a)}(T) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$

x7 $\quad \mathbb{T}_{1,3}^{(a)}(T) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{2} \end{bmatrix}$

- bra: $\langle p_x|\,,\ \langle p_y|\,,\ \langle p_z|$
- ket: $|p_x\rangle\,,\ |p_y\rangle\,,\ |p_z\rangle$

x8 $\quad \mathbb{Q}_0^{(a)}(A) = \begin{bmatrix} \frac{\sqrt{3}}{3} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{3} & 0 \\ 0 & 0 & \frac{\sqrt{3}}{3} \end{bmatrix}$

x9 $\quad \mathbb{Q}_{2,1}^{(a)}(E) = \begin{bmatrix} -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{3} \end{bmatrix}$

x10 $\quad \mathbb{Q}_{2,2}^{(a)}(E) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$

x11 $\mathbb{Q}^{(a)}_{2,1}(T) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$

x12 $\mathbb{Q}^{(a)}_{2,2}(T) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & 0 & 0 \\ \frac{\sqrt{2}}{2} & 0 & 0 \end{bmatrix}$

x13 $\mathbb{Q}^{(a)}_{2,3}(T) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

x14 $\mathbb{M}^{(a)}_{1,1}(T) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{2} \\ 0 & \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$

x15 $\mathbb{M}^{(a)}_{1,2}(T) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{2} & 0 & 0 \end{bmatrix}$

x16 $\mathbb{M}^{(a)}_{1,3}(T) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} & 0 \\ \frac{\sqrt{2}i}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Cluster SAMB

- Site cluster

** Wyckoff: `4a`

y1 $\mathbb{Q}^{(s)}_{0}(A) = \left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$

y2 $\mathbb{Q}^{(s)}_{1,1}(T) = \left[\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}\right]$

y3 $\mathbb{Q}^{(s)}_{1,2}(T) = \left[\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right]$

y4 $\mathbb{Q}^{(s)}_{1,3}(T) = \left[\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}\right]$

- Bond cluster

** Wyckoff: `12a@12b`

y5 $\mathbb{Q}^{(b)}_{0}(A) = \left[\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}\right]$

y6 $\mathbb{T}^{(b)}_{0}(A) = \left[\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}\right]$

y7 $\mathbb{Q}^{(b)}_{2,1}(E) = \left[-\frac{5\sqrt{2778}}{1389}, -\frac{5\sqrt{2778}}{1389}, -\frac{5\sqrt{2778}}{1389}, -\frac{5\sqrt{2778}}{1389}, -\frac{23\sqrt{2778}}{5556}, -\frac{23\sqrt{2778}}{5556}, -\frac{23\sqrt{2778}}{5556}, -\frac{23\sqrt{2778}}{5556}, \frac{43\sqrt{2778}}{5556}, \frac{43\sqrt{2778}}{5556}, \frac{43\sqrt{2778}}{5556}, \frac{43\sqrt{2778}}{5556}\right]$

y8 $\mathbb{Q}^{(b)}_{2,2}(E) = \left[\frac{11\sqrt{926}}{926}, \frac{11\sqrt{926}}{926}, \frac{11\sqrt{926}}{926}, \frac{11\sqrt{926}}{926}, -\frac{21\sqrt{926}}{1852}, -\frac{21\sqrt{926}}{1852}, -\frac{21\sqrt{926}}{1852}, -\frac{21\sqrt{926}}{1852}, -\frac{\sqrt{926}}{1852}, -\frac{\sqrt{926}}{1852}, -\frac{\sqrt{926}}{1852}, -\frac{\sqrt{926}}{1852}\right]$

y9 $\mathbb{T}^{(b)}_{2,1}(E) = \left[-\frac{5\sqrt{2778}i}{1389}, -\frac{5\sqrt{2778}i}{1389}, -\frac{5\sqrt{2778}i}{1389}, -\frac{5\sqrt{2778}i}{1389}, -\frac{23\sqrt{2778}i}{5556}, -\frac{23\sqrt{2778}i}{5556}, -\frac{23\sqrt{2778}i}{5556}, -\frac{23\sqrt{2778}i}{5556}, \frac{43\sqrt{2778}i}{5556}, \frac{43\sqrt{2778}i}{5556}, \frac{43\sqrt{2778}i}{5556}, \frac{43\sqrt{2778}i}{5556}\right]$

y10 $\mathbb{T}^{(b)}_{2,2}(E) = \left[\frac{11\sqrt{926}i}{926}, \frac{11\sqrt{926}i}{926}, \frac{11\sqrt{926}i}{926}, \frac{11\sqrt{926}i}{926}, -\frac{21\sqrt{926}i}{1852}, -\frac{21\sqrt{926}i}{1852}, -\frac{21\sqrt{926}i}{1852}, -\frac{21\sqrt{926}i}{1852}, -\frac{\sqrt{926}i}{1852}, -\frac{\sqrt{926}i}{1852}, -\frac{\sqrt{926}i}{1852}, -\frac{\sqrt{926}i}{1852}\right]$

y11 $\mathbb{Q}^{(b)}_{1,1}(T) = \left[\frac{5\sqrt{2}}{16}, -\frac{5\sqrt{2}}{16}, -\frac{5\sqrt{2}}{16}, \frac{5\sqrt{2}}{16}, \frac{\sqrt{2}}{8}, \frac{\sqrt{2}}{8}, -\frac{\sqrt{2}}{8}, -\frac{\sqrt{2}}{8}, \frac{\sqrt{6}}{16}, -\frac{\sqrt{6}}{16}, \frac{\sqrt{6}}{16}, -\frac{\sqrt{6}}{16}\right]$

y12 $\mathbb{Q}^{(b)}_{1,2}(T) = \left[\frac{\sqrt{6}}{16}, -\frac{\sqrt{6}}{16}, \frac{\sqrt{6}}{16}, -\frac{\sqrt{6}}{16}, \frac{5\sqrt{2}}{16}, -\frac{5\sqrt{2}}{16}, -\frac{5\sqrt{2}}{16}, \frac{5\sqrt{2}}{16}, \frac{\sqrt{2}}{8}, \frac{\sqrt{2}}{8}, -\frac{\sqrt{2}}{8}, -\frac{\sqrt{2}}{8}\right]$

y13 $\mathbb{Q}^{(b)}_{1,3}(T) = \left[\frac{\sqrt{2}}{8}, \frac{\sqrt{2}}{8}, -\frac{\sqrt{2}}{8}, -\frac{\sqrt{2}}{8}, \frac{\sqrt{6}}{16}, -\frac{\sqrt{6}}{16}, \frac{\sqrt{6}}{16}, -\frac{\sqrt{6}}{16}, \frac{5\sqrt{2}}{16}, -\frac{5\sqrt{2}}{16}, -\frac{5\sqrt{2}}{16}, \frac{5\sqrt{2}}{16}\right]$

y14 $\mathbb{T}^{(b)}_{1,1}(T,a) = \left[\frac{5\sqrt{2}i}{16}, -\frac{5\sqrt{2}i}{16}, -\frac{5\sqrt{2}i}{16}, \frac{5\sqrt{2}i}{16}, \frac{\sqrt{2}i}{8}, \frac{\sqrt{2}i}{8}, -\frac{\sqrt{2}i}{8}, -\frac{\sqrt{2}i}{8}, \frac{\sqrt{6}i}{16}, -\frac{\sqrt{6}i}{16}, \frac{\sqrt{6}i}{16}, -\frac{\sqrt{6}i}{16}\right]$

y15 $\mathbb{T}^{(b)}_{1,2}(T,a) = \left[\frac{\sqrt{6}i}{16}, -\frac{\sqrt{6}i}{16}, \frac{\sqrt{6}i}{16}, -\frac{\sqrt{6}i}{16}, \frac{5\sqrt{2}i}{16}, -\frac{5\sqrt{2}i}{16}, -\frac{5\sqrt{2}i}{16}, \frac{5\sqrt{2}i}{16}, \frac{\sqrt{2}i}{8}, \frac{\sqrt{2}i}{8}, -\frac{\sqrt{2}i}{8}, -\frac{\sqrt{2}i}{8}\right]$

y16 $\mathbb{T}^{(b)}_{1,3}(T,a) = \left[\frac{\sqrt{2}i}{8}, \frac{\sqrt{2}i}{8}, -\frac{\sqrt{2}i}{8}, -\frac{\sqrt{2}i}{8}, \frac{\sqrt{6}i}{16}, -\frac{\sqrt{6}i}{16}, \frac{\sqrt{6}i}{16}, -\frac{\sqrt{6}i}{16}, \frac{5\sqrt{2}i}{16}, -\frac{5\sqrt{2}i}{16}, -\frac{5\sqrt{2}i}{16}, \frac{5\sqrt{2}i}{16}\right]$

y17 $\mathbb{T}^{(b)}_{1,1}(T,b) = \left[\frac{5\sqrt{821}i}{3284}, -\frac{5\sqrt{821}i}{3284}, -\frac{5\sqrt{821}i}{3284}, \frac{5\sqrt{821}i}{3284}, -\frac{11\sqrt{821}i}{821}, -\frac{11\sqrt{821}i}{821}, \frac{11\sqrt{821}i}{821}, \frac{11\sqrt{821}i}{821}, \frac{21\sqrt{2463}i}{3284}, -\frac{21\sqrt{2463}i}{3284}, \frac{21\sqrt{2463}i}{3284}, -\frac{21\sqrt{2463}i}{3284}\right]$

y18 $\mathbb{T}^{(b)}_{1,2}(T,b) = \left[\frac{21\sqrt{2463}i}{3284}, -\frac{21\sqrt{2463}i}{3284}, \frac{21\sqrt{2463}i}{3284}, -\frac{21\sqrt{2463}i}{3284}, \frac{5\sqrt{821}i}{3284}, -\frac{5\sqrt{821}i}{3284}, -\frac{5\sqrt{821}i}{3284}, \frac{5\sqrt{821}i}{3284}, -\frac{11\sqrt{821}i}{821}, -\frac{11\sqrt{821}i}{821}, \frac{11\sqrt{821}i}{821}, \frac{11\sqrt{821}i}{821}\right]$

y19 $\mathbb{T}^{(b)}_{1,3}(T,b) = \left[-\frac{11\sqrt{821}i}{821}, -\frac{11\sqrt{821}i}{821}, \frac{11\sqrt{821}i}{821}, \frac{11\sqrt{821}i}{821}, \frac{21\sqrt{2463}i}{3284}, -\frac{21\sqrt{2463}i}{3284}, \frac{21\sqrt{2463}i}{3284}, -\frac{21\sqrt{2463}i}{3284}, \frac{5\sqrt{821}i}{3284}, -\frac{5\sqrt{821}i}{3284}, -\frac{5\sqrt{821}i}{3284}, \frac{5\sqrt{821}i}{3284}\right]$

y20 $\mathbb{M}^{(b)}_{1,1}(T) = \left[-\frac{43\sqrt{4926}i}{13136}, \frac{43\sqrt{4926}i}{13136}, \frac{43\sqrt{4926}i}{13136}, -\frac{43\sqrt{4926}i}{13136}, \frac{25\sqrt{4926}i}{6568}, \frac{25\sqrt{4926}i}{6568}, -\frac{25\sqrt{4926}i}{6568}, -\frac{25\sqrt{4926}i}{6568}, \frac{115\sqrt{1642}i}{13136}, -\frac{115\sqrt{1642}i}{13136}, \frac{115\sqrt{1642}i}{13136}, -\frac{115\sqrt{1642}i}{13136}\right]$

y21 $\mathbb{M}^{(b)}_{1,2}(T) = \left[\frac{115\sqrt{1642}i}{13136}, -\frac{115\sqrt{1642}i}{13136}, \frac{115\sqrt{1642}i}{13136}, -\frac{115\sqrt{1642}i}{13136}, -\frac{43\sqrt{4926}i}{13136}, \frac{43\sqrt{4926}i}{13136}, \frac{43\sqrt{4926}i}{13136}, -\frac{43\sqrt{4926}i}{13136}, \frac{25\sqrt{4926}i}{6568}, \frac{25\sqrt{4926}i}{6568}, -\frac{25\sqrt{4926}i}{6568}, -\frac{25\sqrt{4926}i}{6568}\right]$

y22 $\mathbb{M}^{(b)}_{1,3}(T) = \left[\frac{25\sqrt{4926}i}{6568}, \frac{25\sqrt{4926}i}{6568}, -\frac{25\sqrt{4926}i}{6568}, -\frac{25\sqrt{4926}i}{6568}, \frac{115\sqrt{1642}i}{13136}, -\frac{115\sqrt{1642}i}{13136}, \frac{115\sqrt{1642}i}{13136}, -\frac{115\sqrt{1642}i}{13136}, -\frac{43\sqrt{4926}i}{13136}, \frac{43\sqrt{4926}i}{13136}, \frac{43\sqrt{4926}i}{13136}, -\frac{43\sqrt{4926}i}{13136}\right]$

y23 $\mathbb{Q}^{(b)}_{2,1}(T) = \left[-\frac{43\sqrt{4926}}{13136}, \frac{43\sqrt{4926}}{13136}, \frac{43\sqrt{4926}}{13136}, -\frac{43\sqrt{4926}}{13136}, \frac{25\sqrt{4926}}{6568}, \frac{25\sqrt{4926}}{6568}, -\frac{25\sqrt{4926}}{6568}, -\frac{25\sqrt{4926}}{6568}, \frac{115\sqrt{1642}}{13136}, -\frac{115\sqrt{1642}}{13136}, \frac{115\sqrt{1642}}{13136}, -\frac{115\sqrt{1642}}{13136}\right]$

y24 $\mathbb{Q}^{(b)}_{2,2}(T) = \left[\frac{115\sqrt{1642}}{13136}, -\frac{115\sqrt{1642}}{13136}, \frac{115\sqrt{1642}}{13136}, -\frac{115\sqrt{1642}}{13136}, -\frac{43\sqrt{4926}}{13136}, \frac{43\sqrt{4926}}{13136}, \frac{43\sqrt{4926}}{13136}, -\frac{43\sqrt{4926}}{13136}, \frac{25\sqrt{4926}}{6568}, \frac{25\sqrt{4926}}{6568}, -\frac{25\sqrt{4926}}{6568}, -\frac{25\sqrt{4926}}{6568}\right]$

y25 $\mathbb{Q}^{(b)}_{2,3}(T) = \left[\frac{25\sqrt{4926}}{6568}, \frac{25\sqrt{4926}}{6568}, -\frac{25\sqrt{4926}}{6568}, -\frac{25\sqrt{4926}}{6568}, \frac{115\sqrt{1642}}{13136}, -\frac{115\sqrt{1642}}{13136}, \frac{115\sqrt{1642}}{13136}, -\frac{115\sqrt{1642}}{13136}, -\frac{43\sqrt{4926}}{13136}, \frac{43\sqrt{4926}}{13136}, \frac{43\sqrt{4926}}{13136}, -\frac{43\sqrt{4926}}{13136}\right]$

y26 $\mathbb{Q}^{(b)}_{3,1}(T,1) = \left[\frac{5\sqrt{821}}{3284}, -\frac{5\sqrt{821}}{3284}, -\frac{5\sqrt{821}}{3284}, \frac{5\sqrt{821}}{3284}, -\frac{11\sqrt{821}}{821}, -\frac{11\sqrt{821}}{821}, \frac{11\sqrt{821}}{821}, \frac{11\sqrt{821}}{821}, \frac{21\sqrt{2463}}{3284}, -\frac{21\sqrt{2463}}{3284}, \frac{21\sqrt{2463}}{3284}, -\frac{21\sqrt{2463}}{3284}\right]$

y27 $\mathbb{Q}^{(b)}_{3,2}(T,1) = \left[\frac{21\sqrt{2463}}{3284}, -\frac{21\sqrt{2463}}{3284}, \frac{21\sqrt{2463}}{3284}, -\frac{21\sqrt{2463}}{3284}, \frac{5\sqrt{821}}{3284}, -\frac{5\sqrt{821}}{3284}, -\frac{5\sqrt{821}}{3284}, \frac{5\sqrt{821}}{3284}, -\frac{11\sqrt{821}}{821}, -\frac{11\sqrt{821}}{821}, \frac{11\sqrt{821}}{821}, \frac{11\sqrt{821}}{821}\right]$

y28 $\mathbb{Q}^{(b)}_{3,3}(T,1) = \left[-\frac{11\sqrt{821}}{821}, -\frac{11\sqrt{821}}{821}, \frac{11\sqrt{821}}{821}, \frac{11\sqrt{821}}{821}, \frac{21\sqrt{2463}}{3284}, -\frac{21\sqrt{2463}}{3284}, \frac{21\sqrt{2463}}{3284}, -\frac{21\sqrt{2463}}{3284}, \frac{5\sqrt{821}}{3284}, -\frac{5\sqrt{821}}{3284}, -\frac{5\sqrt{821}}{3284}, \frac{5\sqrt{821}}{3284}\right]$

Site and Bond

Table 5: Orbital of each site

#	site	orbital
1	Co	$\|s\rangle$, $\|p_x\rangle$, $\|p_y\rangle$, $\|p_z\rangle$

Table 6: Neighbor and bra-ket of each bond

#	head	tail	neighbor	head (bra)	tail (ket)
1	Co	Co	[1]	[s,p]	[s,p]

Site in Unit Cell

Sites in (conventional) cell (no plus set), SL = sublattice

Table 7: 'Co' (#1) site cluster (4a), .3.

SL	position ($\boldsymbol{s}$)	mapping
1	[0.14000, 0.14000, 0.14000]	[1,5,9]
2	[0.36000, 0.86000, 0.64000]	[2,7,12]
3	[0.86000, 0.64000, 0.36000]	[3,8,10]
4	[0.64000, 0.36000, 0.86000]	[4,6,11]

Bond in Unit Cell

Bonds in (conventional) cell (no plus set): tail, head = (SL, plus set), (N)D = (non)directional (listed up to 5th neighbor at most)

Table 8: 1-th 'Co'-'Co' [1] (#1) bond cluster (12a@12b), D, $|\boldsymbol{v}|$= 0.61384 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[0.22000,-0.28000,-0.50000]	[0.25000, 0.00000, 0.89000]	[1]	(2,1)	(1,1)	[0,1,1]
2	[-0.22000, 0.28000,-0.50000]	[0.25000, 0.00000, 0.39000]	[2]	(1,1)	(2,1)	[0,-1,0]
3	[-0.22000,-0.28000, 0.50000]	[0.75000, 0.50000, 0.61000]	[3]	(4,1)	(3,1)	[0,0,0]
4	[0.22000, 0.28000, 0.50000]	[0.75000, 0.50000, 0.11000]	[4]	(3,1)	(4,1)	[0,0,-1]
5	[-0.50000, 0.22000,-0.28000]	[0.89000, 0.25000, 0.00000]	[5]	(4,1)	(1,1)	[1,0,1]
6	[-0.50000,-0.22000, 0.28000]	[0.39000, 0.25000, 0.00000]	[6]	(1,1)	(4,1)	[0,0,-1]
7	[0.50000,-0.22000,-0.28000]	[0.61000, 0.75000, 0.50000]	[7]	(3,1)	(2,1)	[0,0,0]
8	[0.50000, 0.22000, 0.28000]	[0.11000, 0.75000, 0.50000]	[8]	(2,1)	(3,1)	[-1,0,0]

continued ...

Table 8

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
9	[-0.28000,-0.50000, 0.22000]	[0.00000, 0.89000, 0.25000]	[9]	(3,1)	(1,1)	[1,1,0]
10	[0.28000,-0.50000,-0.22000]	[0.00000, 0.39000, 0.25000]	[10]	(1,1)	(3,1)	[-1,0,0]
11	[-0.28000, 0.50000,-0.22000]	[0.50000, 0.61000, 0.75000]	[11]	(2,1)	(4,1)	[0,0,0]
12	[0.28000, 0.50000, 0.22000]	[0.50000, 0.11000, 0.75000]	[12]	(4,1)	(2,1)	[0,-1,0]